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Testing technology and quality control

Lecture Production of electric motors and machines
Vorlesung Produktion elektrischer Motoren und Maschinen

The lecture unit "Testing technology and quality control" provides an overview of the causes of failures of electrical drives as well as suitable test procedures.

Electrical machine testing ensures quality production and trouble-free operation throughout the product lifecycle.

After attending the lecture unit "Testing technology and quality control" you should:

- **know the causes of failure of electrical machines and their effects,**
- **know the different measurement methods and**
- **in particular have internalized the underlying measurement principles and**
- **decide which test method should be used for a predefined machine and operating parameters**

Graphic: Omron

Graphic: ZF Friedrichshafen

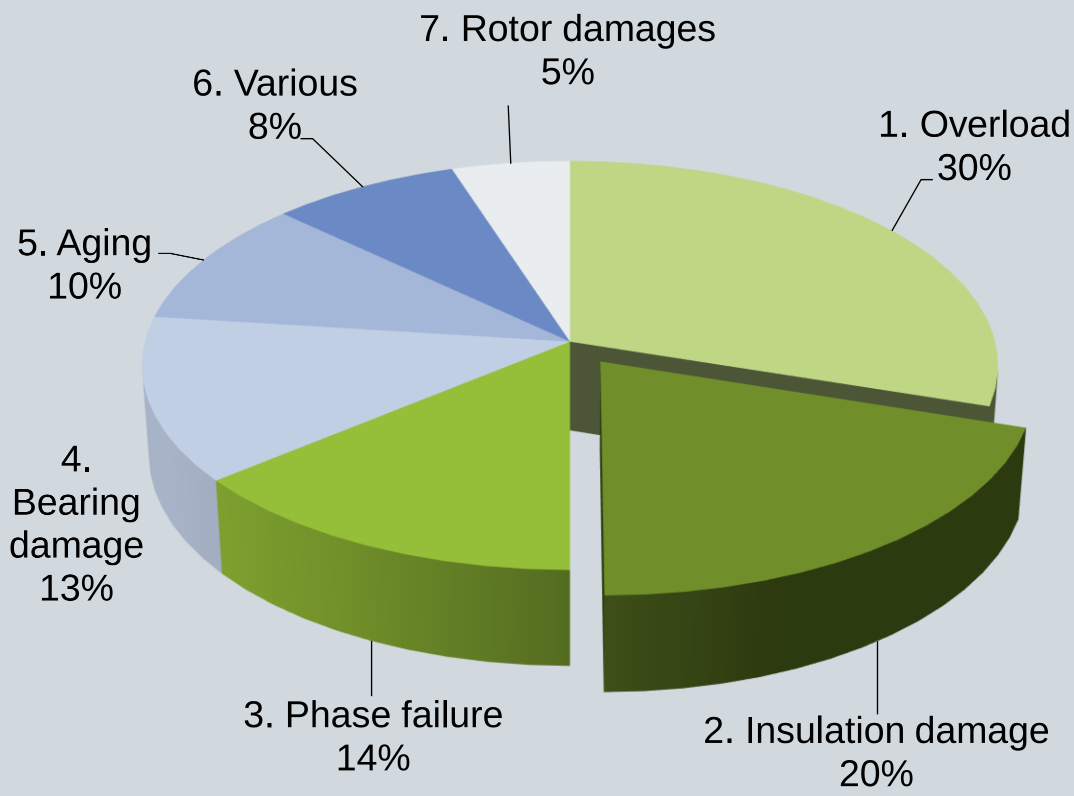
Graphic: Siemens

Graphic: Kuka

The lecture unit "Testing technology and quality control" provides an overview of the causes of failures of electrical drives as well as suitable test procedures.

- **Causes of failure and degradation mechanisms of the dielectrics of motors and windings**
- Test methods for detecting defects in the insulation system
- Characterization of soft magnetic materials
- Characterization of hard magnetic materials

The causes of failure in electric motors are not exclusively related to insulation damage in the windings.



Source: Rockwell Automation

1. Overload
2. Insulation damage
   a. Winding short circuit: Short circuit between individual windings of a winding
   b. Winding short circuit: Short circuit between individual windings
   c. Phase short circuit: Short circuit between individual phases
3. Phase failure (e.g. interruption of phase supply lines)
4. Bearing damage
5. Ageing (need for action can be derived from trend measurements)
6. Various (operation outside specifications, lightning strike)
7. Rotor damage

The causes of insulation damage are varied and can also overlap.

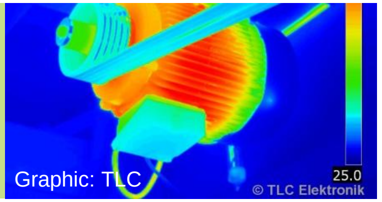
Slide from Lec. 8


Graphic: sr.wiki.com

Overvoltage

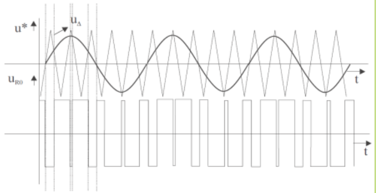
Electrical breakdown when the limiting field strength of the dielectric is exceeded

Thermal Load



Graphic: TLC

Thermal runaway due to insufficient heat dissipation



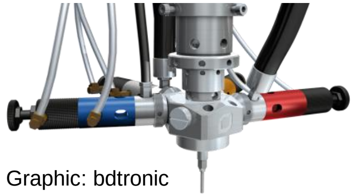
PWM operation

High pulse frequency: Voltage gradients cause polarization losses and large potential differences

Environment variables

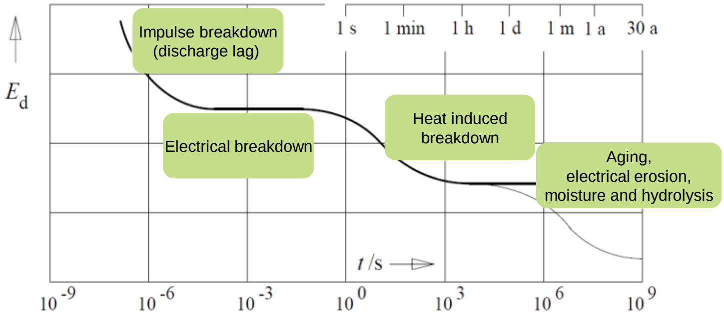

Graphic: 123RF

Chemicals, dirt, abrasion, moisture


Graphic: bdtronic

Processing

Mechanical stress, e.g. from winding process, cracks/defects: Ignition source for partial discharge erosion resulting in failure



Source: Küchler

Failure mechanisms of a solid dielectric depending on exposure time and field strength

An increased thermal load on the insulation material leads to a reduction in the remaining service life.

IEC (DIN EN) 60034-1:

Classification of winding insulation according to maximum permissible continuous temperature in insulation material classes

Insolation class	Maximum permissible temperature [°C]	
Y	90	
A	105	
E	120	
B	130	Relevant Classes f. E-motors
F	155	Relevant Classes f. E-motors
H	180	Relevant Classes f. E-motors
C	> 180	

Lecture unit 8

Arrhenius law for failure rates as a function of temperature:

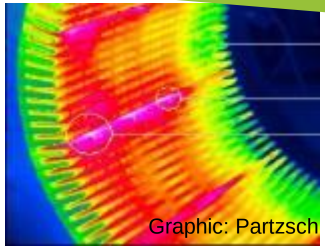
$$r(T_1) = r(T_2)e^{\frac{E_A}{k_B}\left(\frac{1}{T_1}-\frac{1}{T_2}\right)}$$

Operating the windings 10°C above the allowable continuous temperature will cut the remaining life in half.

$$P_{zu} = P_\delta + P_I \ \text{mit}\ P_\delta = U^2 \cdot \omega C \cdot \tan\delta_U e^{\beta(T-T_U)}$$

with:
- P_{zu} – Supplied power loss
- P_δ – Power loss in the dielectric
- P_I – Current heat output

$\tan\delta$	$e^{\beta(T-T_U)}$	$P_\delta = \sum_{i=0}^{\infty}(\tan\delta)_i(\omega_i C U_i^2)$	$P_I = R * I^2$	$T - T_U$
▪ $\tan\delta$: Dielectric dissipation factor ▪ High loss factor in the dielectric $\cong 10^{-6} \ldots 10^{-2}$ ▪ Depending on material and purity	▪ β: Coefficient ▪ Over-proportional increase in thermal losses ▪ Frequency dependence on $\tan(\delta)$	▪ **Harmonic components** ▪ ω: Frequency ▪ C: Capacitance of the dielectric ▪ $_i$: Order of the harmonic ▪ High-frequency control	▪ External heat supply (ambient temperature, neighboring conductors) ▪ Ohmic losses	▪ T: Temperature ▪ T_U: Temp. of environment ▪ Insufficient heat dissipation ▪ Low thermal conductivity ▪ Contamination


Graphic: Partzsch

The power loss delivered to the winding is made up of the ohmic losses and the losses in the dielectric.

Source: "Hochspannungstechnik", A. Küchler; Springer Verlag

The insulation system can be damaged and aged by various processes. Aging results in reduced electrical strength.

Feed-out radius and angle during needle winding and wire tension forces

+

Friction during the drawing-in process

+

Torsion of the wire during flyer winding

Voltage breakdown strength of winding wire after the winding process only approx. 60% of as-delivered condition

Curing processes for insulation:

By energizing and heating the windings, the dielectric of the insulating varnish is thermally pre-aged.

Material aging:

Mechanical, chemical, thermal and electrical stress, weather, operating environment, radiation

Embrittlement, cracks

Electrochemical ageing:

Electrochemical changes due to the combination of humid atmosphere and electric field

Electrochemical breakdown

Electrothermal ageing:

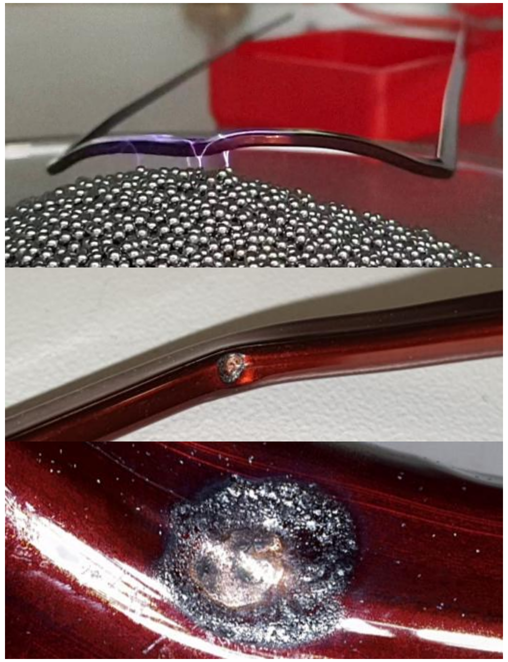
Degradation reactions in the insulation material due to dielectric and ohmic heat losses

Degradation of the insulation

Partial discharge ageing:

Internal partial discharge or surface discharge due to local field strength increase at imperfections in the insulation

Partial discharge erosion

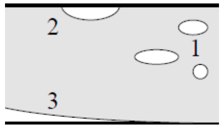


Preventive measures in production and product design

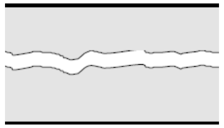
Reduction of influences on insulation

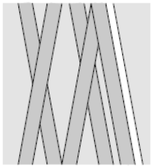
The formation of erosion channels, so-called "trees" due to partial discharges can lead to an electrical breakdown.

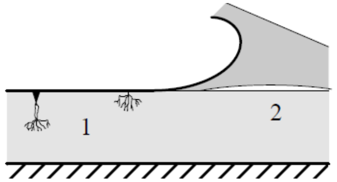
- Partial discharges are caused by local field strength increases due to field displacement or conductive peaks
- Reduction in electrical strength due to
  - Inclusions or bubbles
  - Cracks due to mechanical processing or manufacturing defects



Cavities, voids without (1) and with (2) electrode contact, and detachments (3)

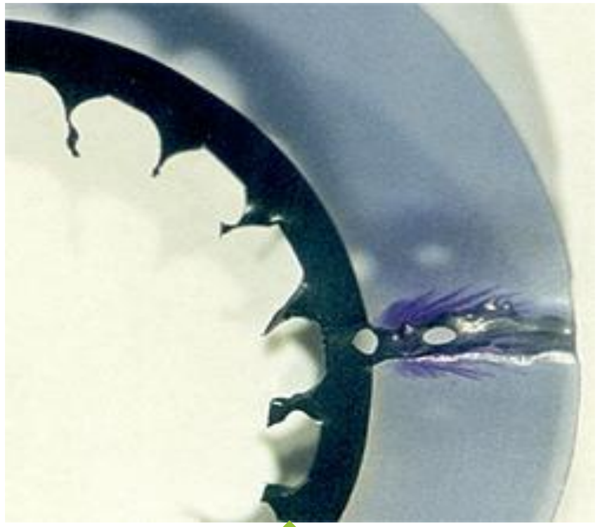

Cracks, gaps, and imperfect coatings or impregnations


Fiber interface delamination (GRP)



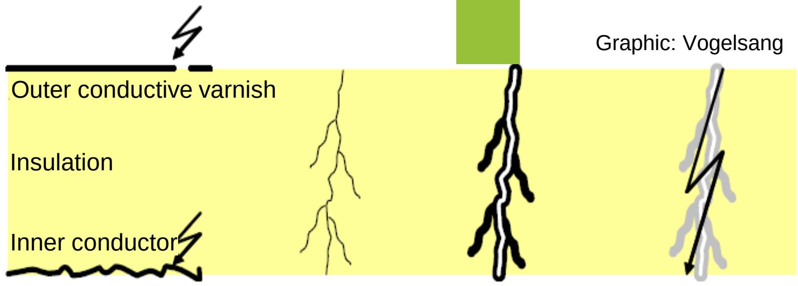
Cable with end seal:(1) voids due to treeing(2) voids at the interface

Graphics: Küchler


Graphic: Vogelsang

- Partial discharges: Partial discharges often do not immediately reduce electrical strength
- For solid organic insulators, partial discharge erosion will reduce the life of the insulation system
- Progressive erosion leads to erosion breakthrough
  - "Treeing" describes processes during erosion
  - The time between "tree start" and final breakthrough depends on several parameters

Graphic: Vogelsang



Tree start | Tree growth | Widening of the tree canals | Final breakthrough

Time

The lecture unit "Testing technology and quality control" provides an overview of the causes of failures of electrical drives as well as suitable test procedures.

- Causes of failure and degradation mechanisms of the dielectrics of motors and windings
- **Test methods for detecting defects in the insulation system**
  - High-voltage tests
  - Surge voltage test
  - Partial discharge measurement
  - Polarization index PI and tanδ
  - Temperature measurement
  - Power measurement
- Characterization of soft magnetic materials
- Characterization of hard magnetic materials

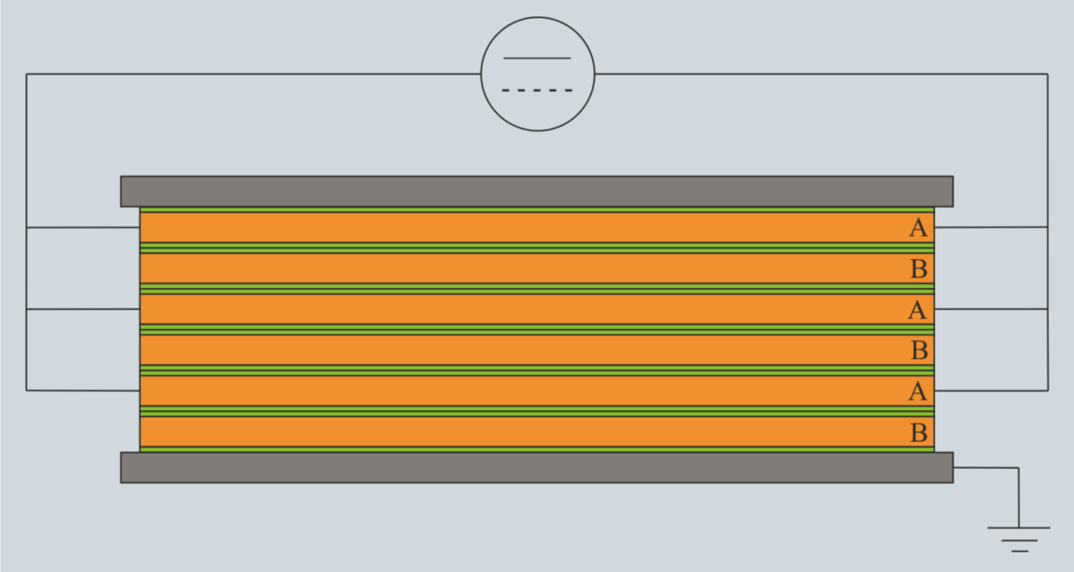
Various measurements are used to check the insulation properties of wound stators.

Slide from Lec. 8

- **Insulation resistance measurement**
- **High voltage test**
- Protective conductor resistance measurement
- Resistance measurement
- **Surge voltage test**
- **Partial discharge measurement**
- Polarization index measurement
- Loss angle measurement
- Temperature measurement/ thermography
- Performance test with built-in rotor

Measuring duration

Test voltage



Winding → ⚡ ← **Winding**

Winding → ⚡ ← **Sheet package**

Measuring current → **Insulation resistance**

- e.g. 500 V DC direct voltage according to DIN VDE 0100-600
- No influence with DC voltage due to small capacitances of the insulation material
- For large machines (large capacities), longer measurement is still necessary to ensure that the high voltage is present
- High insulation resistance is an indication of good insulating capacity

The high voltage (HV) test with DC voltage is usually an overload test.
The HV test with AC voltage is carried out in the same way as the measurement with DC voltage.

FAPS

Parameters for high voltage measurement

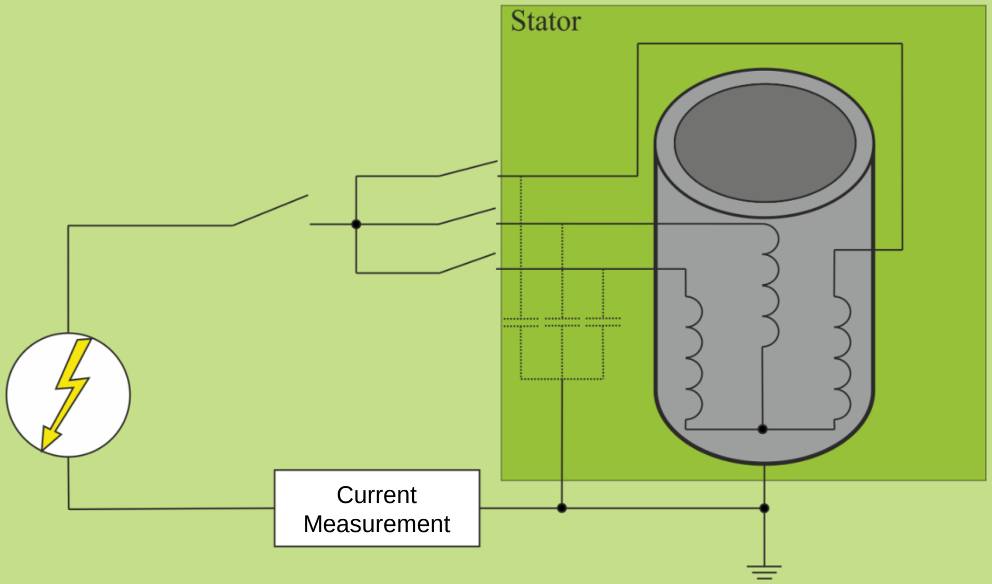
Test duration | Min. current | Max. current | Voltage | Rise time | Max. Capacity | Total current

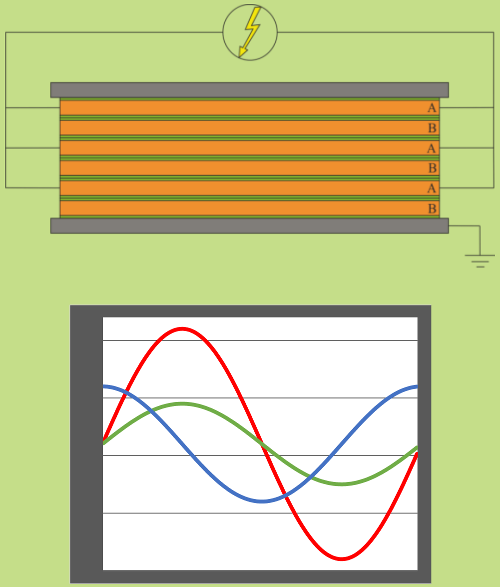
Winding – **Winding**

Winding – **Iron core**

Winding – **T-sensor**

T-sensor – **Iron core**





Measured variables

Measuring current (with AC) | Test voltage | Parasitic capacity | Total current

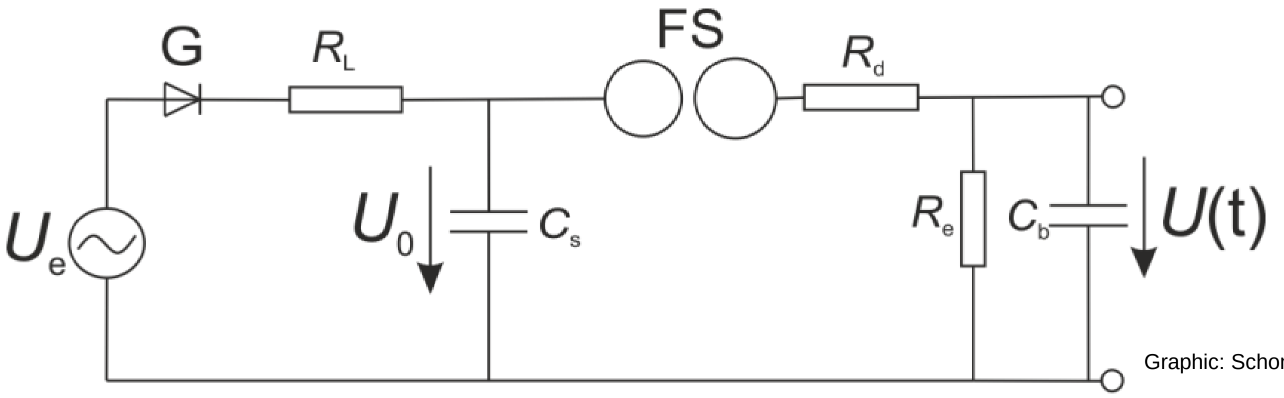
FAPS

The surge test simulates surges caused by external influences or transient switching within the network.

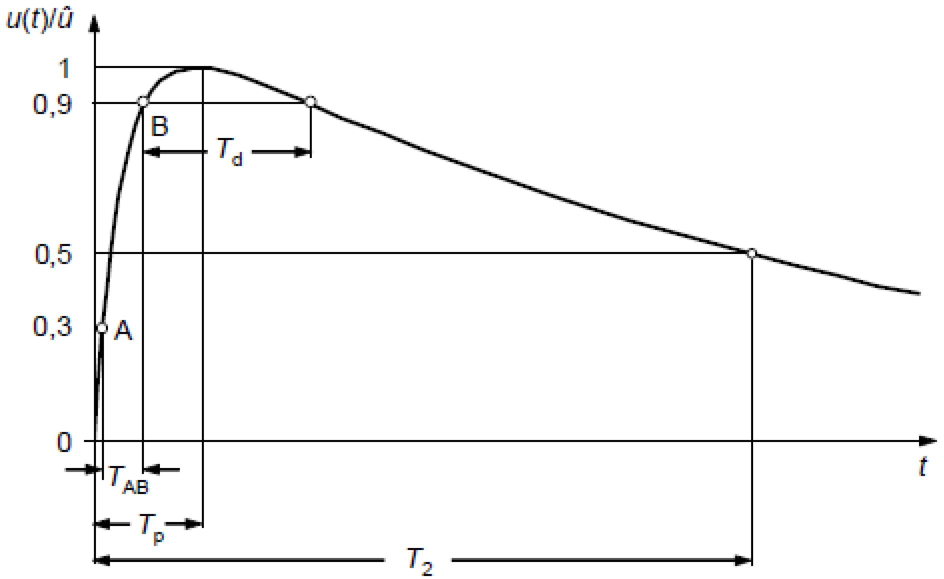
Charging the surge capacitor C_s until charging voltage U_0 is reached

Pulse-like discharge of C_s via spark gap FS

Tapping the test voltage via load capacitor C_b



Graphic: Schon



Graphic: Schon

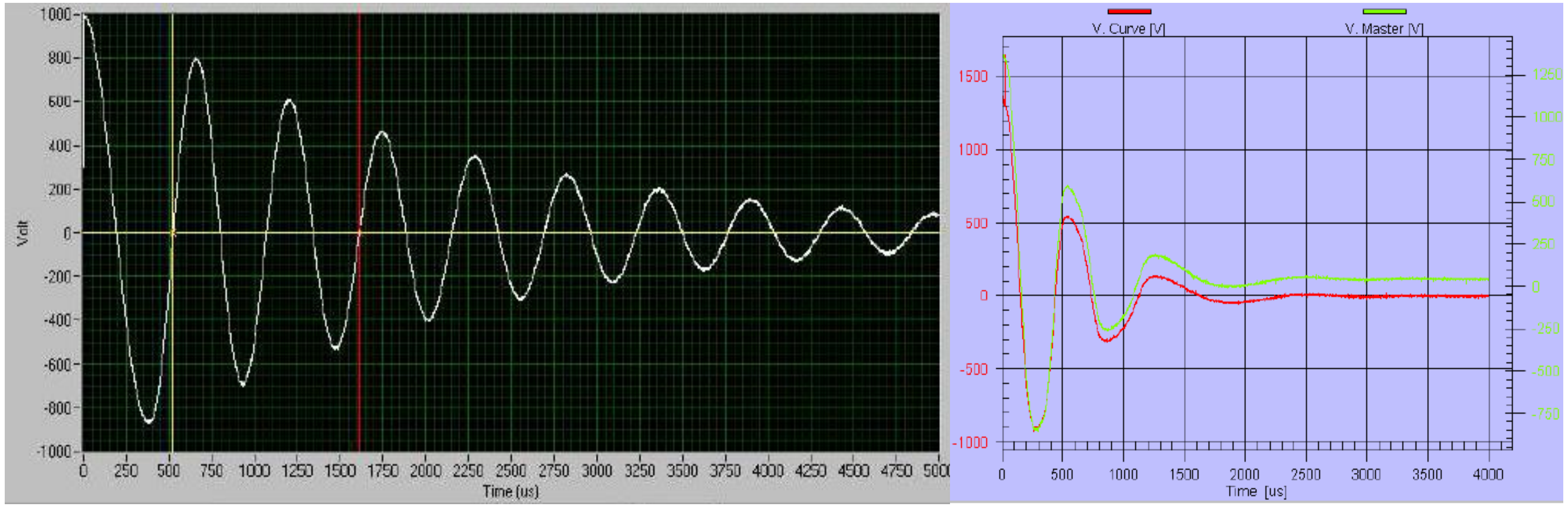
Standardized form in IEC 60060

- Peak time T_p
- Back half-life T_2
- T_d is the time in which the voltage is greater than 0.9 û
- T_{AB}, time between 0.3 û and 0.9 û
- Forehead time $T_1 = \frac{1}{0{,}6} T_{AB}$
- Differentiation between lightning and switching impulse voltages via end time: For switching impulse voltages $T_1 > 20\mu s$
- Damping resistor R_d limits the surge time Surge voltage curve
- Discharge resistance R_e influences back half-life

The typical step response curve of a surge excited winding exhibits PT2 behavior.

Shock excitation → **Test item** → **Step response**:

$$a(t) = K - \frac{K}{\sqrt{1-d^2}} e^{-\frac{dt}{T}} \cdot \sin\left[\frac{\sqrt{1-d^2}}{T} t + \tan^{-1}\frac{\sqrt{1-d^2}}{d}\right]$$



- The curve of the measured voltage after surge voltage excitation corresponds to that of a PT2 element with periodic damping D<1
- With the damping factor $\boldsymbol{D} = \frac{\boldsymbol{R}}{2\sqrt{\frac{L}{C}}}$ of an L-C-R resonant circuit and the time constant T and the amplification factor K

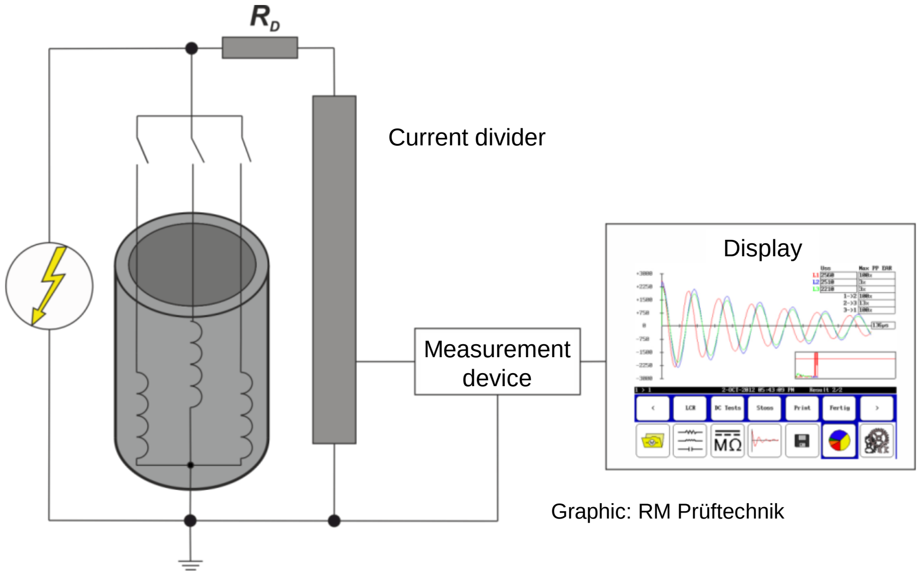
Modern surge testers contain all the necessary components and provide information on several characteristics of the winding material under test in a single measurement.

Voltage generation

Voltage divider - resistive or capacitive

Measuring device - surge voltmeter or digital recorder

Display device - e.g. oscilloscope, monitor



Graphic: RM Prüftechnik

Rapid change in voltage over time → **Shock wave through windings of a winding leads to potential differences** → **E-field causes punctures** → **Defects visible during step response** → **Comparison with reference specifications**

Create a master curve with the desired parameters

Tolerance band for various parameters (amplitude ⇒ damping, frequency ⇒ inductance, capacitance)

Surge voltage curve must be within the positive and negative tolerance curves resulting from the master curve

Short-circuited windings, deviation in number of windings

Partial contacts with (partial) discharges

Corona discharges (surface discharges)

Increased parasitic capacitance, resistance

Deviation of the inductance

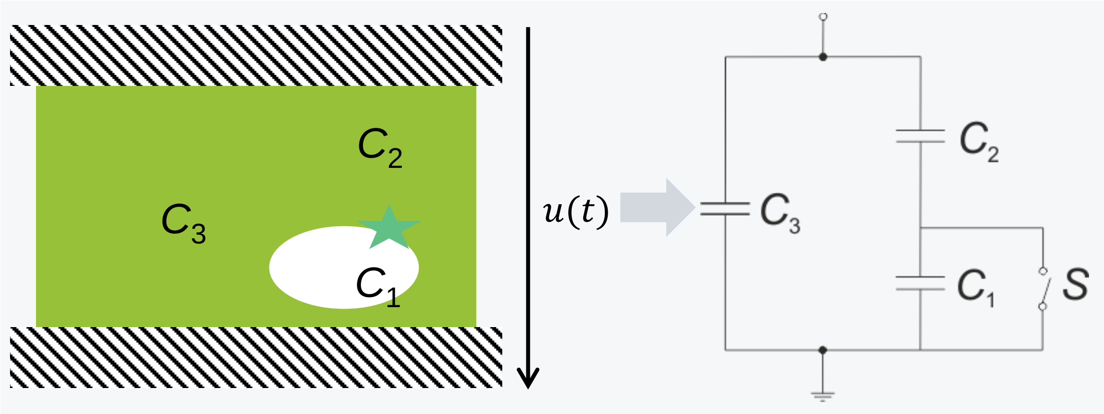
Basic considerations of the partial discharge pulses in cavities can be carried out using a simplified equivalent circuit diagram.

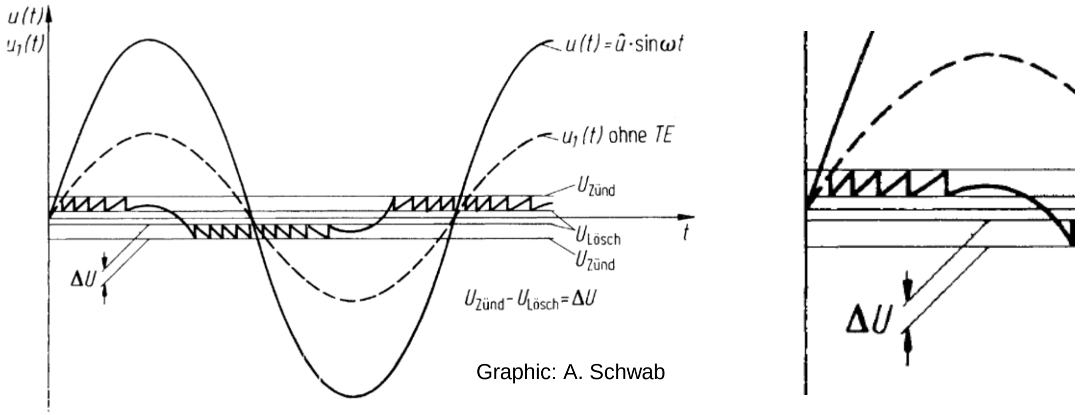
Voltage applied:
$u(t) = \hat{u} \cdot \sin \omega t$

Cavity stress without PD:
$$\boldsymbol{u_1(t) = u(t)\frac{C_2}{C_1 + C_2}}$$

Exceeding a critical ignition voltage

Discharge takes place in cavity capacity C_1





Graphic: A. Schwab

$u_1(t) > u_Z$ as well as the presence of a starting electron

Independent ionization processes in the dielectric cavity

Voltage dip at the cavity. When falling below the extinguishing voltage u_L → Charging of the cavity

- $\boldsymbol{\frac{du(t)}{dt} = max.} \rightarrow$ **Internal partial discharges**
- $\boldsymbol{\frac{du(t)}{dt} = 0} \rightarrow$ **Surface discharges**

Recharging of the cavity can be recorded as a current pulse on a partial discharge measuring device

Several components are required for partial discharge (PD) measurement in accordance with IEC 60270. The coupling quadripole (CQp) couples the signal to the PD measuring device.

Harmonic-free AC voltage source (50 Hz) to supply the circuit

PD-free transformer to generate the necessary test voltage

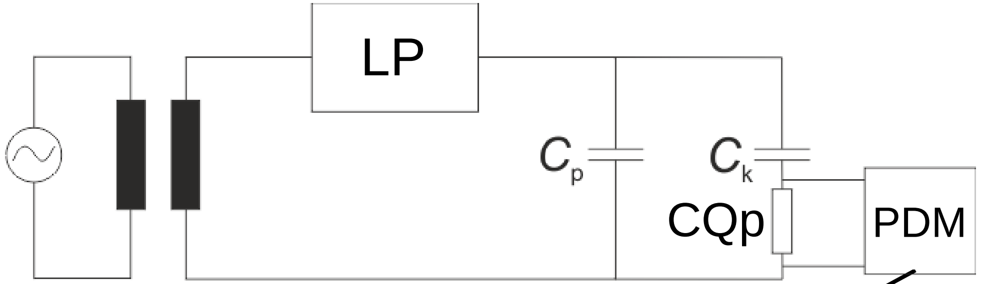
Low-pass (LP) circuit for interference suppression of mains transients

Test dielectric C_P in the form of the insulating system of a wound stator

Partial discharge-free coupling capacitor C_k : supplies current pulses after PD events

Coupling quadripole (CQp) for decoupling the measuring signal

Partial discharge meter (PDM) with integrator



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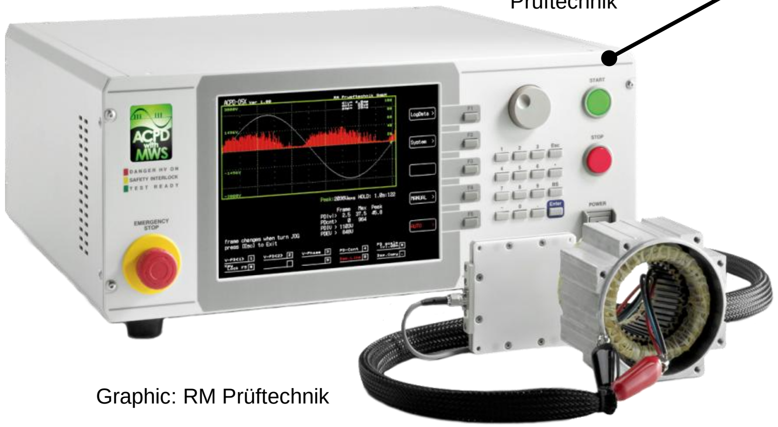


Graphic: RM Prüftechnik

Connection to the test object is usually made using antennas or connection terminals

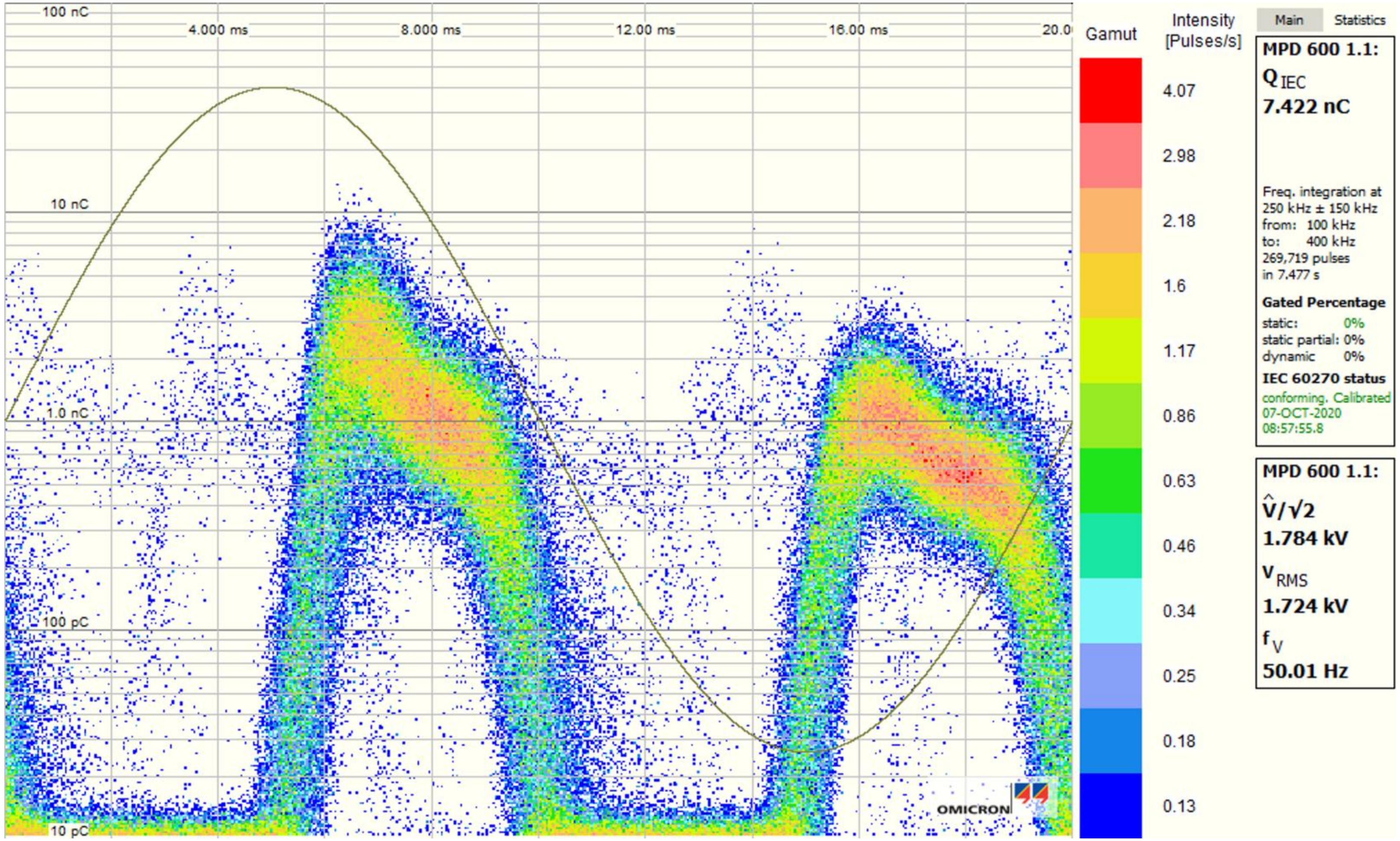
CQp ideally matched impedance Z_M , not a single concentrated component but often a network;
Broadband measurement: e.g. ohmic resistance R_M (matching the characteristic impedance of the measuring cable, usually 50Ω);
Narrow-band measurement: e.g. tuned resonant circuit

Prevention of possible reflections at impedance jumps

Suppression of standing waves on the measuring line → otherwise distortion of the measuring signal

A typical representation of static partial discharge (PD) events is the plot of the magnitude of the PD, incidence rate of the PD, at the point in time over a period in a histogram.

- Abscissa (x-axis): One period (usually at 50 Hz)
- Ordinate (y-axis): Magnitude of the PD in pC or nC
- Color value: Intensity of the PD at the same period point and with the same magnitude



To estimate the erosive effect of partial discharges, the partial discharge energy W_{TE} and the discharge frequency are considered.

With each PD pulse, the apparent charge
$\boldsymbol{Q = C \cdot \Delta U}$
is resupplied subsequently

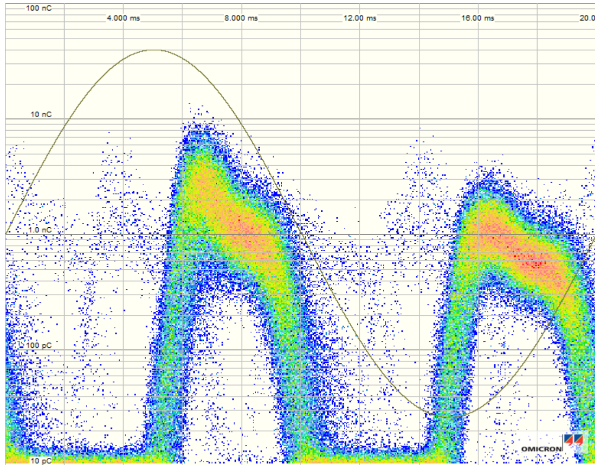
$\boldsymbol{Q}$ is much smaller than the charge actually transferred $\boldsymbol{Q}_1$

The actual charge $\boldsymbol{Q}_1$ has not been measurable to date

Cannot be calculated analytically due to the unknown type, position and geometry of the cavities

Energy conversion corresponds to the energy stored in the cavity. With complete discharge, the following relationship results:

$$W_{TE} \approx \frac{1}{2} C_h \Delta uh^2 = \frac{1}{2} \Delta Q \Delta u_h$$

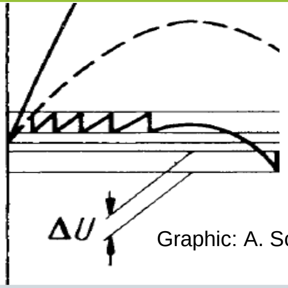


In addition to PD intensity, PD incidence is also important for partial discharge erosion

$\Delta Q = Q_s \cdot \frac{C_h}{C_s}$ and

$\Delta u_h = \Delta U_{n,n+1} \cdot \frac{C_s}{C_h}$

Q_s and ΔU_{n+1} can be measured at the terminals and thus recorded externally



Graphic: A. Schwab

Amount of charge converted per unit of time $N \cdot Q_s$ leads to total energy $N \cdot W_{TE}$

The capacitive division ratio is shortened and the converted partial discharge energy can thus be measured:

$$W_{TE} \approx \frac{1}{2} \cdot Q_s \cdot \Delta U_{n,n+1}$$

Permissible energy conversion during partial discharges constant → Lower partial discharge intensity at high operating voltages

Eroding effect due to many small as well as a few large partial discharges

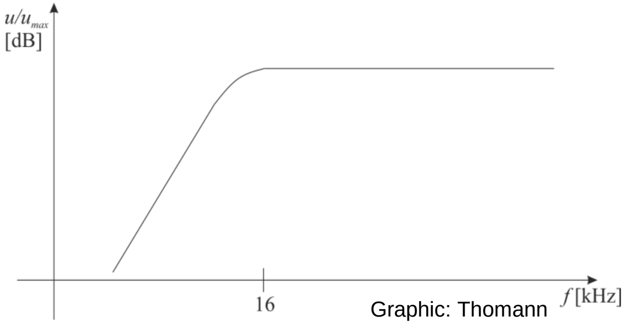
In sensitive partial discharge measurement, interference affects the quality of the measurement result.

Electromagnetic radiation	Shielding of the measuring room	 Graphic: h+h High Voltage Gm
Network-related interference, e.g. switching pulses, harmonics	Current limiting resistors, winding inductance and capacitance	
Radio interferers, e.g. radio transmitters, W-LAN, GPRS, UMTS, LTE	Narrow-band PD measuring devices	 Graphic: RM Prüftechnik
Contact noise	Defined conductor connections, e.g. by screwing or wedging	
Partial discharges in the measurement setup	Rounded edges, shielding hoods	 Graphic: Neonela Edelstahl

In addition to electrical partial discharge detection optical and acoustic detection methods are used.

Type	Detectable effects	Interferences	Measurement
Optical	▪ Precise localization of PD in transparent dielectrics ▪ External corona discharges detectable with residual light amplifiers	▪ Glare effects from extraneous light, e.g. daylight ▪ Ambient lighting in the laboratory or production hall	▪ Complete dimming ▪ Filtering the UV light from the discharges
Acoustic	Localization using time-of-flight measurement with structure-borne sound microphones and directional microphones	▪ Sound reflections ▪ Material-related differences in running time ▪ Ambient noise	Narrow frequency bands in the ultrasonic range ($f > 16\ kHz$)

Graphic: Thomann

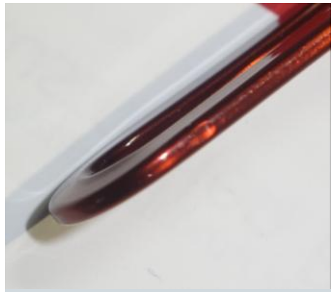


Graphic: Thomann

Various methods are available for qualifying the electrical properties and bending quality.

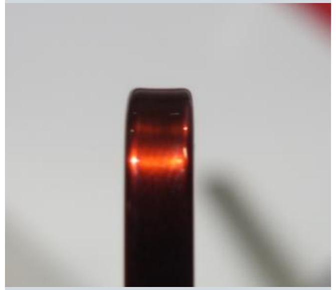
Slide from VE6

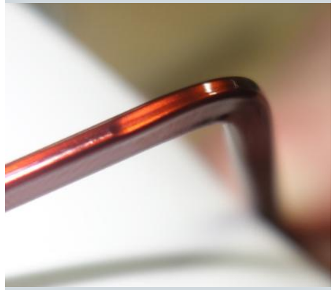

Protrusion, bulge


Dents


Flattening in the bend


Radius deviation

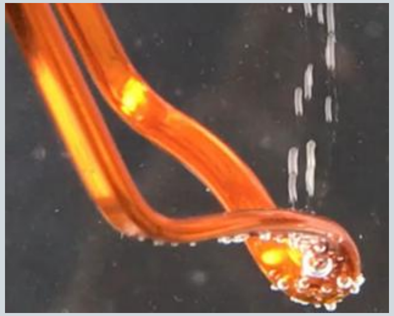

Sinking


Flattening and material postponement

Pin-hole test

also: error count test

- Electrolysis bath for the visualization of damage to the insulation layer due to rising gas bubbles
- Copper wire as anode and aluminum sheet as cathode
- Standard: DIN EN 60851-5



Durability

- Immersion of the bent copper wire samples in a grist bath
- Up to 5.5 kV AC voltage (60 Hz)
- Standard: DIN EN 60851-5

Partial discharge measure-ment

- Defined placement of a bent wire sample on a steel electrode
- Tracing of a voltage curve and recording of PDIV (inception voltage of the PD) and PDEV (drop-out voltage of the PD)
- Up to 5.5 kV AC voltage (60 Hz), no standardization

The precise measurement of ohmic resistances enables the detection of minimal deviations in electrical conductors.

Lecture unit 7

FAPS

- Detection of faulty hot crimp connections of phases U, V, W and star points
- Detection of conductor interruptions
- Detection of changes in the ohmic resistance of the phases U, V, W due to material or assembly problems
- Detection of resistance differences between coils

Two-wire measurement technology for resistors

Analog meters, e.g. moving-coil meters, round-coil moving-iron meters, electrodynamic meters

Digital multimeter with stabilized constant current source

Advantages of four-wire measurement technology in bridge circuit:

Measurement errors due to contact resistances are avoided.
Division into low-impedance supply and high-impedance measuring branch

Four-wire measurement technology for very small resistances

Analog measuring methods, e.g. Thompson bridge or Kelvin bridge
(resistance measuring bridge, similar to the wheatstone bridge) up to $10^{-7}\Omega$

Digital measuring instruments with stabilized constant current source, reference resistor and differential amplifier

The lecture unit "Testing technology and quality control" provides an overview of the causes of failures of electrical drives as well as suitable test procedures.

- Causes of failure and degradation mechanisms of the dielectrics of motors and windings
- Test methods for detecting defects in the insulation system
- **Characterization of soft magnetic materials**
- Characterization of hard magnetic materials

As part of the product test, a holistic evaluation of the insulation layer is performed under various stresses.

Measurement of the insulation resistance:

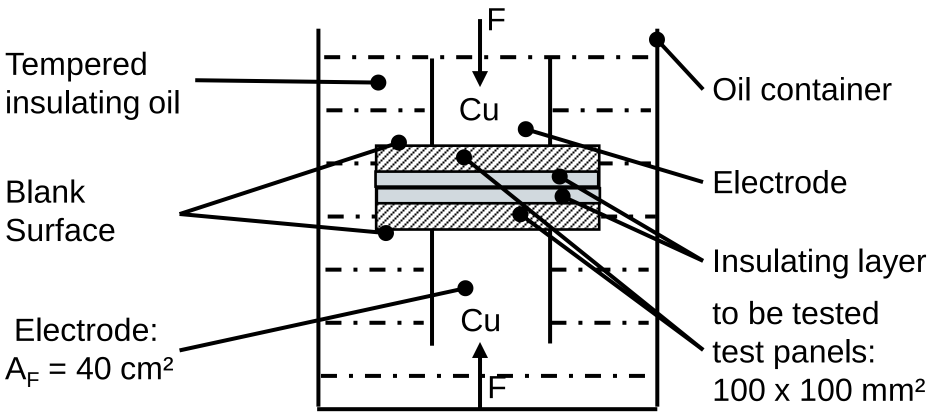
- Testing with applied pressure and temperature stress
  - Current is measured between two layers of insulation
  - Two sheets are used, with the insulation layer being removed from each side.
  - Sample geometry: 100 mm x 100 mm, Two sheets are used, with the insulation layer being removed from each side.
  - Resistance is calculated from current and voltage measurement
  - The quality criterion is the specific surface resistance
  - Two sheets are used, with the insulation layer being removed from each side.
- Procedure according to Franklin (next slide)
  - Current is measured through an insulating layer with ten electrodes connected in parallel
  - Determination of the surface resistance
  - Quality control of insulating coatings

Impermeability of the insulating layer (copper test)

Adhesive strength of the insulating layer

Source: Tzscheutschler

Pressure and temperature



Franklin Tester

Graphic: Brockhaus

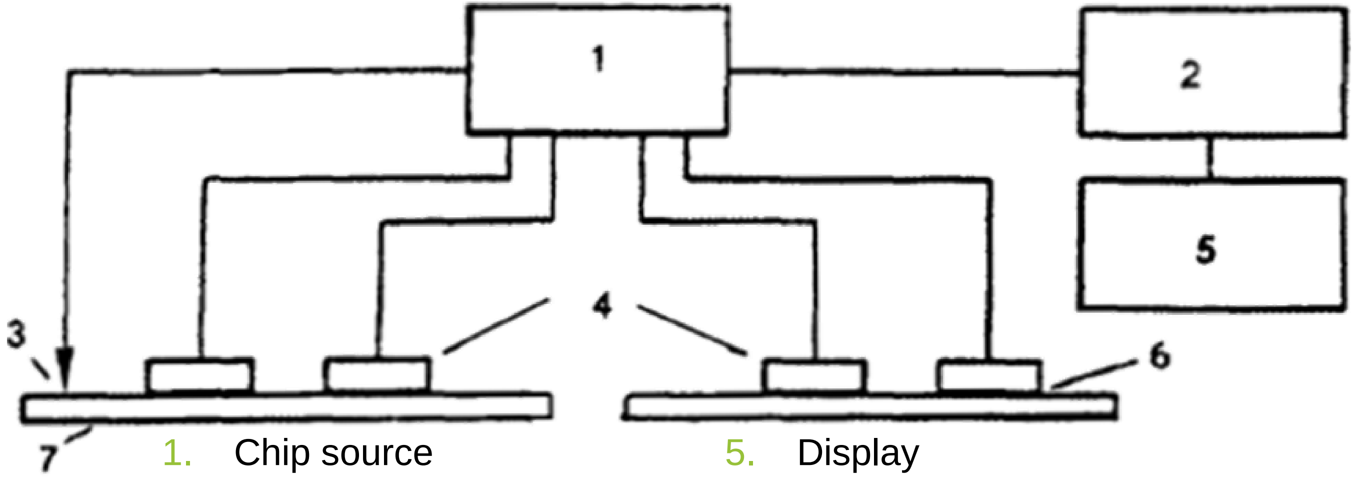
FAPS

The surface resistance of the sheet metal insulation can be be measured with a "Franklin tester" in accordance with DIN EN 10282.

Important key data for measurement:

- Arrangement of the tester as shown in Figure
- Applying 10 metallic flat contacts with a defined surface to the coated / insulated surface of the sheet metal
- Clamping a test specimen between the plate and the electrode arrangement
- Use of a DC voltage source that maintains a stabilized voltage of 500 mV across the electrodes up to 2.5 A
- After calibrating the system and reading the corresponding currents for different electrode arrangements, the test values **C** of the insulation resistance can be calculated from the measured currents.

Arrangement of the tester



1. Chip source
2. Microprocessor
3. Twist drill electrode
4. Contact electrode (10x)
5. Display
6. Insulating coating
7. Test specimen

Graphic and source: DIN 10282

$$C = A\left[\frac{U}{\frac{1}{10}\sum_1^{10} I_A} - \frac{R}{10}\right] = 645\left[\frac{0{,}5}{\frac{1}{10}\sum_1^{10} I_A} - 0{,}5\right]$$

with:

- C - Coefficient of the surface resistance [Ωmm²]
- A - Total area of the 10 contact electrodes [mm²]
- U - Voltage applied to the electrodes and 5 Ω resistors [V]
- R - resistance connected in series with each electrode, equal to 5 Ω
- I_A - measured total electrode current (10 values) [A]

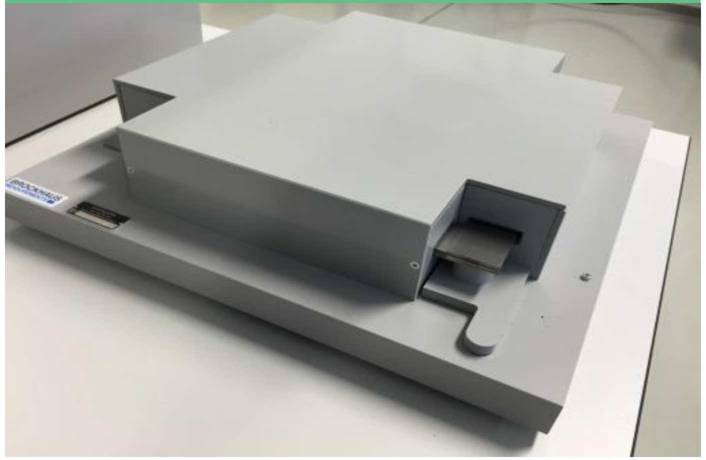
Soft magnetic materials metrology includes measurements for incoming inspection, magnetic properties, and production quality.

Single sheet tester

- Suitable for incoming goods inspection and quality control
- For sheets up to 150 mm wide
- Shows the influence of the separation process on the electromagnetic properties
- Measurement according to standard DIN IEC 60404-3

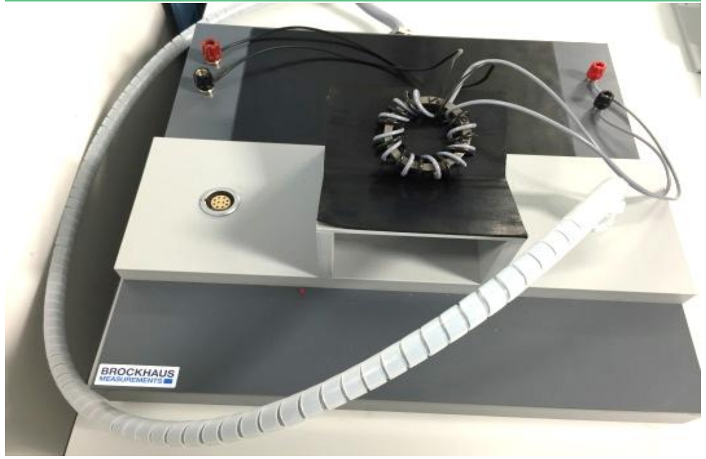
Epstein frame

- Suitable for incoming goods inspection and quality control
- Only for sheets with $30\ \mathrm{mm} \pm 0.2\ \mathrm{mm}$ and $280\ \mathrm{mm} \leq l \leq 320\ \mathrm{mm}$
- Minimum sample mass $240\ \mathrm{g}$ (several probe sheets required)
- Measurement lengthwise and crosswise to the rolling direction
- Measurement according to standard DIN IEC 60404-2



Ring core tester

- Shows the influence of the cutting and joining process on the electromagnetic properties of a sheet metal package
- Manual sample handling, winding and measurement
- Risk of short circuit due to insulation fault
- Measurement according to standard DIN IEC 60404-6



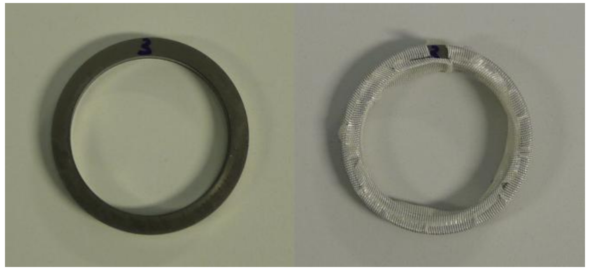
The influence of the joining process on the losses of a stator laminated core can be evaluated with a ring core tester.

Determination of the magnetic characteristics using the ring method:

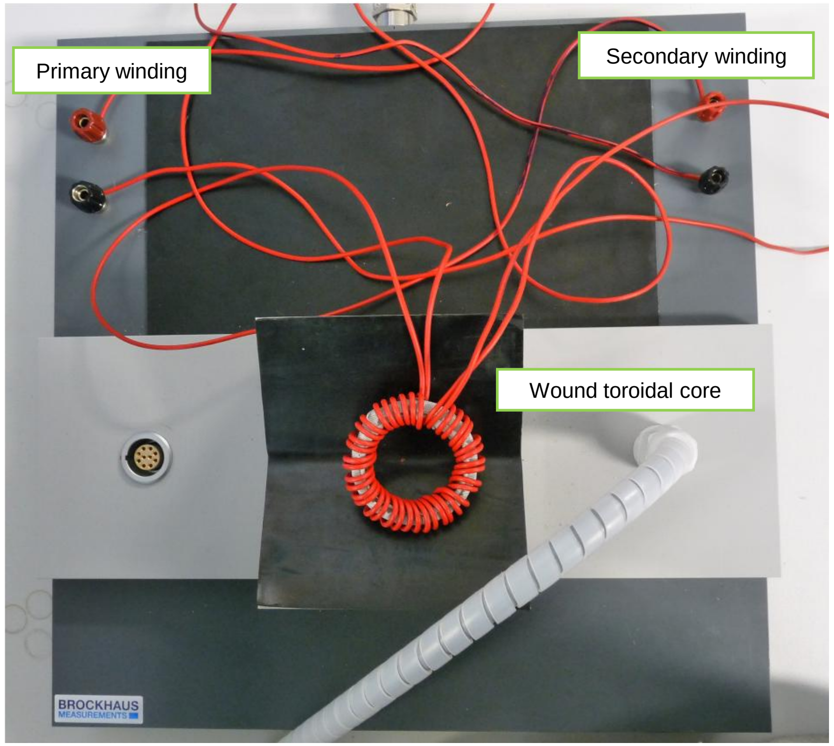
- Determination of commutation curve and hysteresis loop
- Reference sample is a homogeneous, non-welded ring with a rectangular or circular cross-section
- Test samples can feature different joining techniques like welding, gluing, interlocking
- Winding of a secondary winding (for measuring) made of insulated copper wire
- Winding of a magnetization winding to achieve the required maximum magnetic field strength

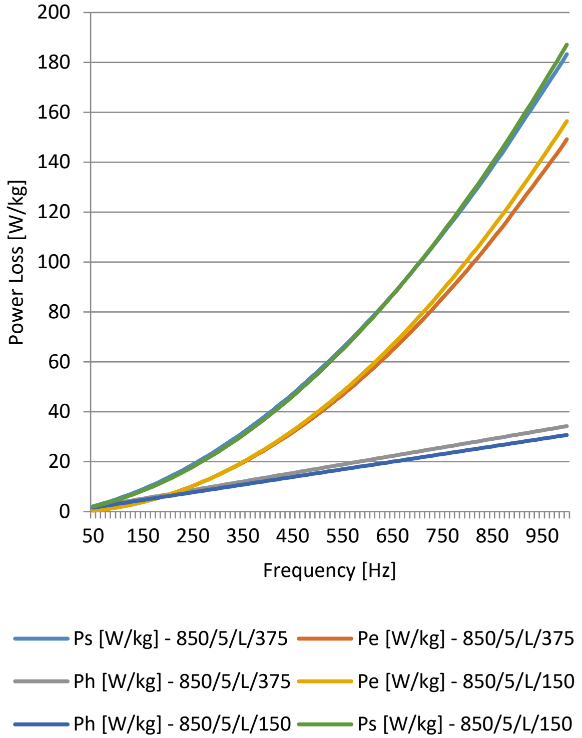
Possible applications:

- Comparison of different packaging methods
- Influence of cutting technologies



Additional insulation around the toroidal core

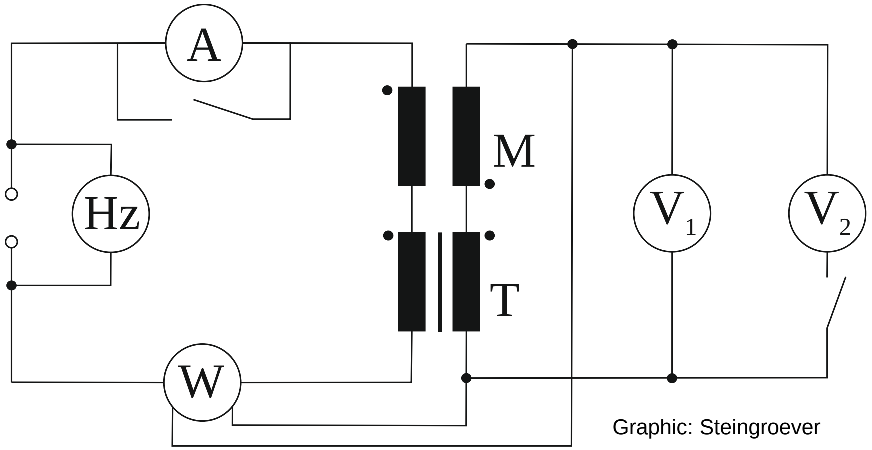




The underlying measurement principle for measurements according to the IEC 60404 series of standards is similar in several respects.

The measuring principle of the Epstein frame, single strip tester and ring core tester is the same:

- Transformer measurement of losses by recording the secondary voltage and power measurement
- Windings are fixed for Epstein frames and single strip testers, variable for the ring core tester
- Compensation of losses in the measuring devices
- Compensation of the air flow through additional inductance
- Calculation of eddy current and hysteresis losses by recording the specific losses at two frequencies

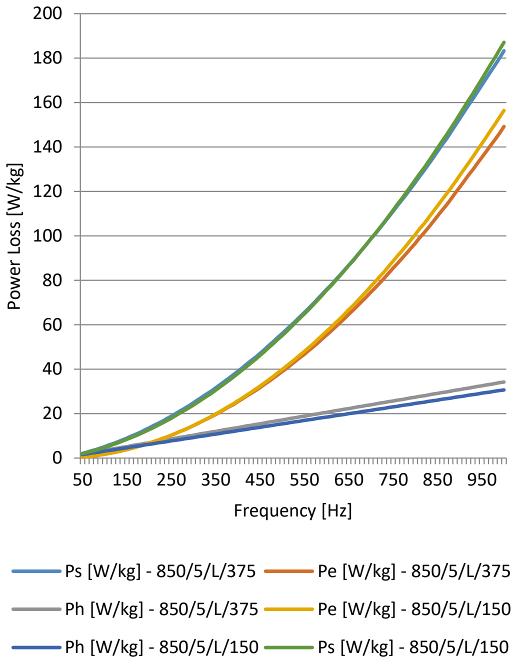


Graphic: Steingroever

$$P = \frac{N_1}{N_2} * \frac{1}{T} * \int_0^T i_1(t) * u_2(t)\, dt$$

$$P_s = \left[P \frac{N_1}{N_2} - \frac{(1{,}111|\overline{U_2}|)^2}{R_i} \right] \frac{l}{m l_m}$$

- The specific losses are made up of
  $P_s = P_e + P_h + (P_{ex})$
- The terms are defined by the following variables:
  $P_e = \frac{\pi^2 \sigma d^2}{6\rho} B^2 f^2$ and $P_h = k \frac{4H_c}{\rho} B f$
- $P_e \sim f^2$; $P_h \sim f$
- Possible separation of P_e and P_h in the test software



DIN EN 10106 and DIN EN 10251 define the test methods for geometric properties and tolerances, but these are usually checked by the manufacturer before delivery.

Thickness (according to DIN EN 10106)

- Thickness measurement at a distance of more than 30 mm from the edges
- Measurement using a micrometer with a resolution of 0.001 mm

Width (according to DIN EN 10106)

Measurement transverse to the longitudinal axis of the product

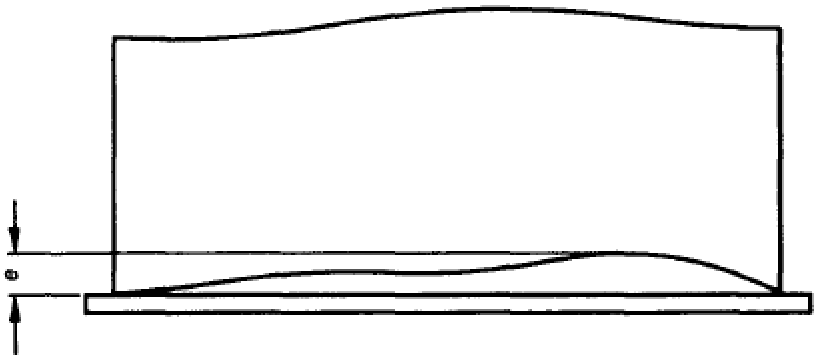
Edge curvature (according to DIN EN 10251)

- Placement of the sample on the support table
- Measurement using a ruler at the extreme point of the concave side
- The largest gap (e) between the edge and the ruler must be measured at its largest point

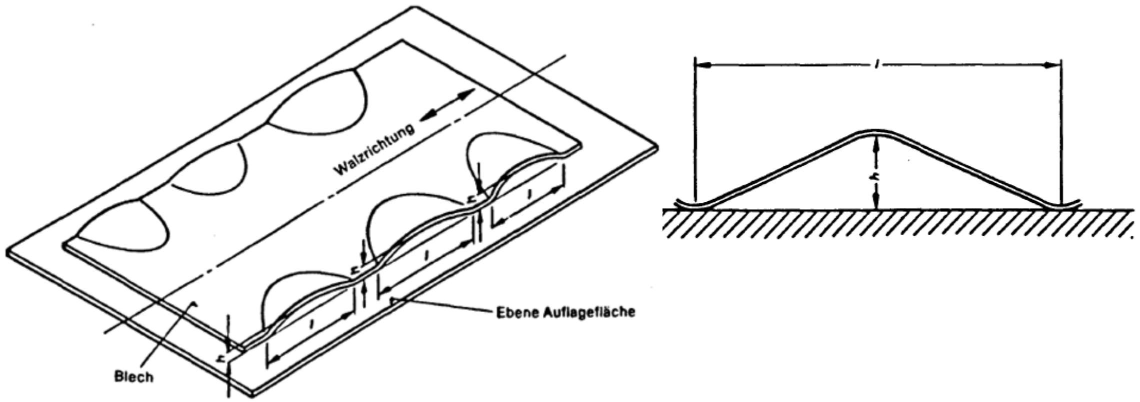
Flatness (waviness factor) (according to DIN EN 10251)

- Place sample on sufficiently large support tables - no overhang over table edges
- The height of the largest elevation (h) and its length (l) must be measured

Edge curvature



Ripple



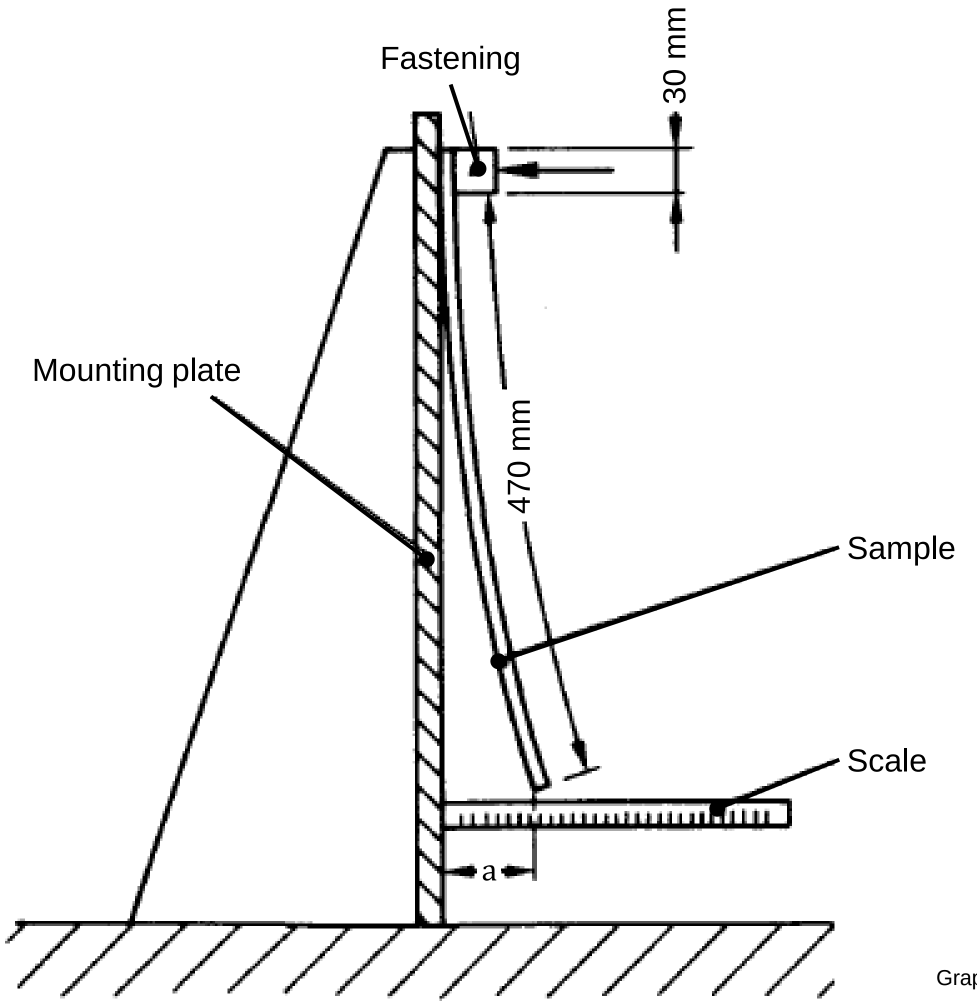
Graphics: DIN 10251

DIN EN 10106 and DIN EN 10251 define the test methods for geometric properties and tolerances, but these are usually checked by the manufacturer before delivery.

Bendiness (DIN EN 10251)

- Sample must consist of a sheet sample of 500 mm length and a width equal to the delivery width of the belt
- Axis of the sample must be parallel to the rolling direction of the sheet
- Vertical mounting of the sample on a mounting plate - the upper edge must be attached to the mounting plate over a height of 30 mm
- The full width of the sample must be in contact with the mounting plate
- Evaluation: measure the distance (a) between the lower edge of the sample and the mounting plate

Bendiness



Graphic: DIN 10251

DIN EN 10106 and DIN EN 10251 define the test methods for geometric properties and tolerances, but these are usually checked by the manufacturer before delivery.

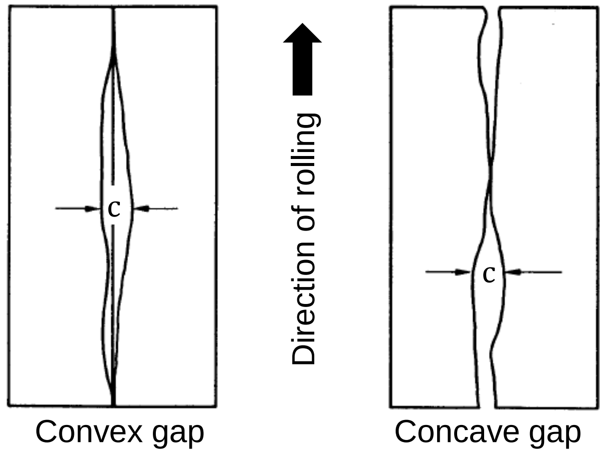
Cutting line deviation due to internal stresses

- According DIN EN 10251
- Cutting the sample along the longitudinal axis
- Weighing down the two parts to ensure evenness
- Bring the cut edges together as close as possible.
- Maximum distance (c) to be measured between the cut edges

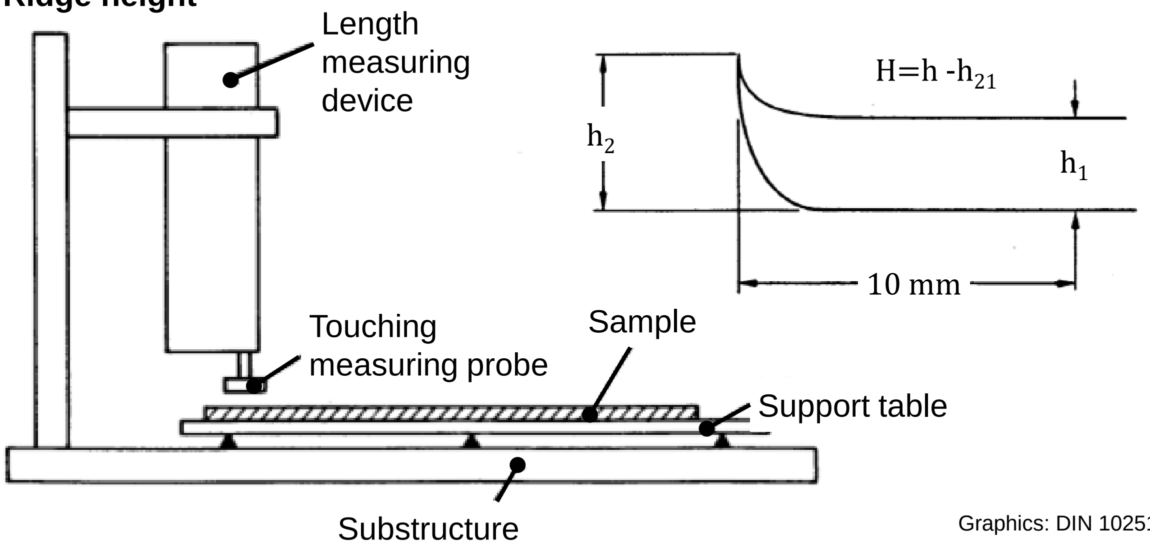
Ridge height

- → Difference between the thickness of the sheet measured at the cut edge and the thickness measured at a distance of 10 mm from this edge
- Measurement of the ridge height with length measuring device, e.g. touching measuring probe
- Movement of the touching measuring probe vertical to the table
- Constant temperature during the measurement

Cutting line deviation



Ridge height



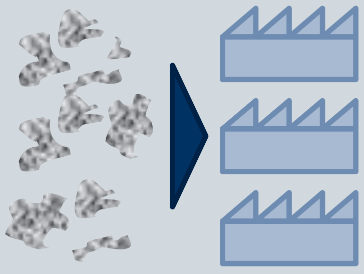
Graphics: DIN 10251

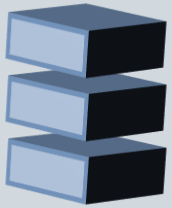
The lecture unit "Testing technology and quality control" provides an overview of the causes of failures of electrical drives as well as suitable test procedures.

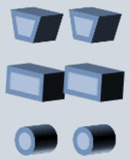
- Causes of failure and degradation mechanisms of the dielectrics of motors and windings
- Test methods for detecting defects in the insulation system
- Characterization of soft magnetic materials
- **Characterization of hard magnetic materials**

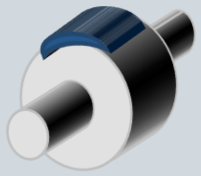
Every process step in the production of hard magnetic materials needs a suitable quality inspection process.

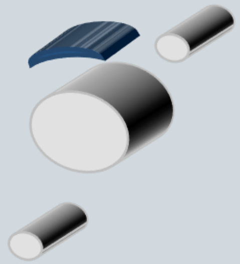
Examination	Examination	Examination	Examination	Examination	Examination
Processing raw materials	**Semi-finished product** *Grünling*	**End-product magnet**	**Automated** Cycle time <5s **Partially automated** Cycle time ~20s	**Motors**	**Disassembly**
Material composition	Magnetic / geometric / mechanical and material properties		Function test / magnetic properties and material defects	Function test	Magnetic properties











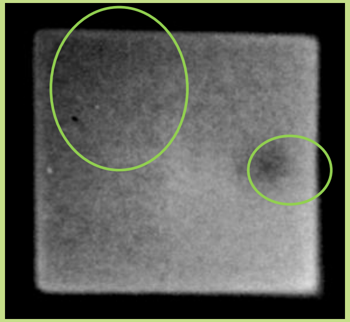
Process sequence for the production of permanently excited rotors:

Raw materials → Processing / semi-finished product → Automated production / assembly → Product → Recycling

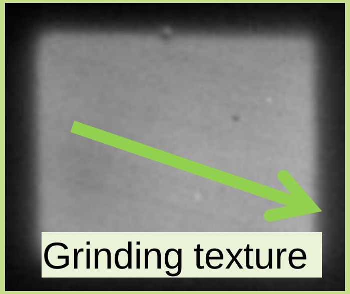
Optical, geometric and magnetic measuring principles are used to detect geometric variables, material defects and magnetic properties.

- Macroscopically, **the surfaces** of permanent magnets are examined for their magnetic structure
- **Cracks**, **scratches** and **surface roughness** have an effect on the field distribution near the surface
- **Stray fields** are formed and can be visualized by corresponding sensors

- An important influence on the energy balance and the torque curve in electric drives is a **homogeneous magnetization**
- **Air inclusions**, **blowholes** and **material defects** in the magnet also have a significant influence on the magnetic field distribution



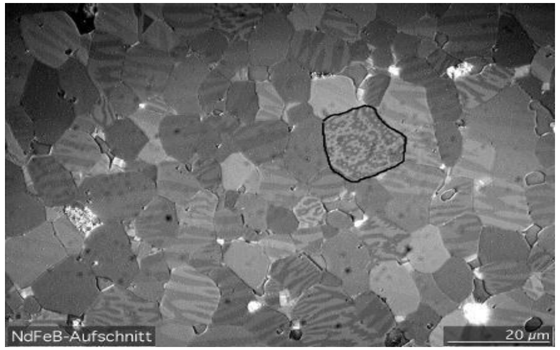
Processing parameters, such as the grinding direction, can also be detected magnetically



Magnets must meet high quality requirements, which are tested using various methods.

Lecture unit 3

Material properties

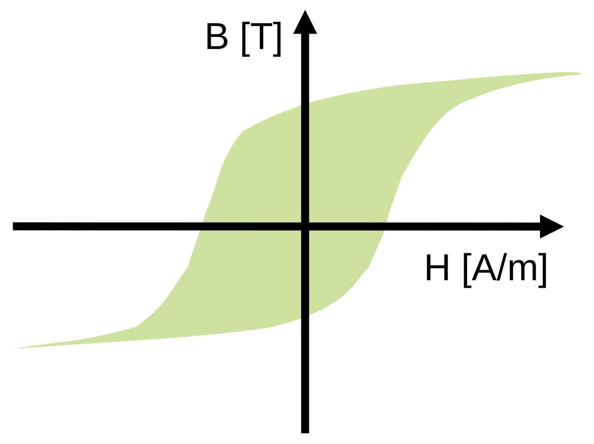


Graphic: ThyssenKrupp Magnettechnik

Stability and robustness

- Temperature resistance
- Corrosion resistance
- Chemical resistance
- Mechanical strength
- Domain structure

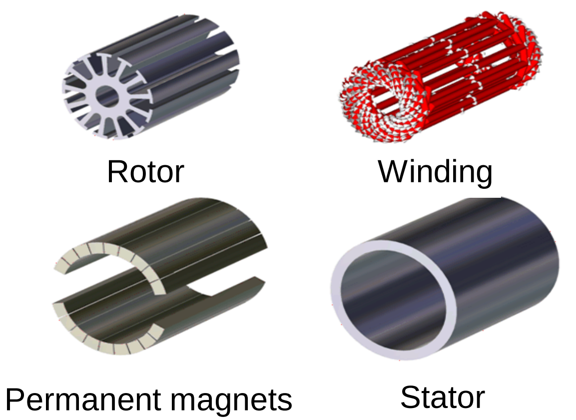
Magnetic properties



Influence on function

- Remanence polarization
- Coercive field strength
- Magnetic field distribution
- Magnetic moment
- Hysteresis

Geometry properties / material defects



Graphic: Swiss Federal Office of Energy SFOE (Switzerland)

Influence of assembly

- Surface finish
- Geometry
- Damage
- Microstructure / Inclusions

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The magnetic central characteristics of hard magnetic materials play a decisive role in the efficiency of electric drives.

Testing the magnetic characteristics		
Remanence	■ Residual magnetism when the magnetic excitation is removed ■ Measurement in hysteresis graphs with low-drift fluxmeters	Decisive for the magnetic power density or motor output
Coercive field strength	■ Demagnetizing field strength ■ Measurement in hysteresis graphs with low-drift fluxmeters	■ Protection against demagnetization or field weakening ■ Target value Demagnetization (recycling)
Field distribution	■ Homogeneity of the field distribution outside and inside the material ■ Hall sensors, gauss meters, flux meters	Decisive for the dynamic properties
Magnetic moment	■ Far field measurement ■ AMR sensors / fluxmeters with Helmholz coil, etc.	Decisive for the magnetic power density

FAPS

Various sensor types allow quality assurance of individual magnets as well as entire magnet systems.

Possible measuring methods and equipment

Single magnet

Far field

- Magnetoresistive sensors
- Magnetic moment coils
- Helmholtz coils

Near field

- Hall effect sensors
- Magneto-optical sensors
- Field coils

Magnetic system

Near field

- Field coils
- Magneto-optical sensors
- Hall effect sensors

In order to evaluate the function of overall systems, the field distribution can be measured around the circumference of the rotors.

Measurement of the field distribution on permanent magnet rotors

Quality criteria of the tested magnetic sample

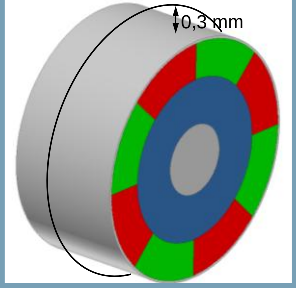
- Field strength
- Resulting preferred direction
- Homogeneous field distribution over the circumference (axial)
- No field disturbance due to material defects, cracks or blowholes

Quality criteria for measuring systems

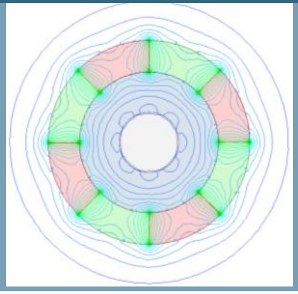
- High measurement dynamics
- Short cycle time / measuring time
- Reproducibility
- Low environmental influence
- Can be automated

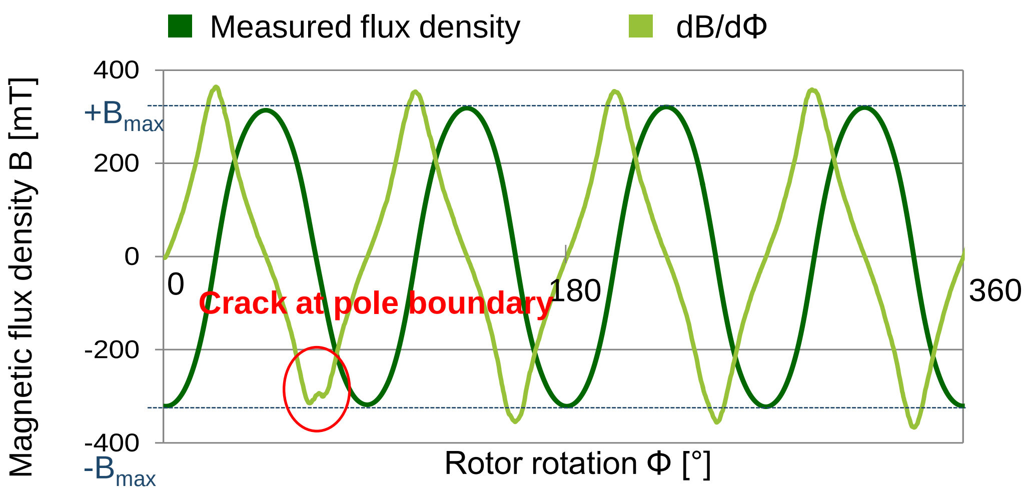
Numerical representation of a homogeneous field distribution by measurement with sensor arrays and evaluation of the sensor data

Analysis model



Field distribution (absolute)



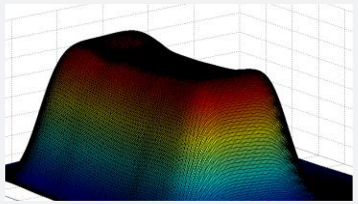


The magnetic moment is determined in the fluxmeter by integrating the voltage induced by the magnet into the Helmholtz coil.

Magnet

- Magn. moment
  $m = \Phi \cdot k$[V s cm]
- Polarization
  $J = \frac{m}{V}$ [T]

V: Volume of the solenoid [cm³]



Fluxmeter

Integration (time wise) of the induced voltage

Simplified:

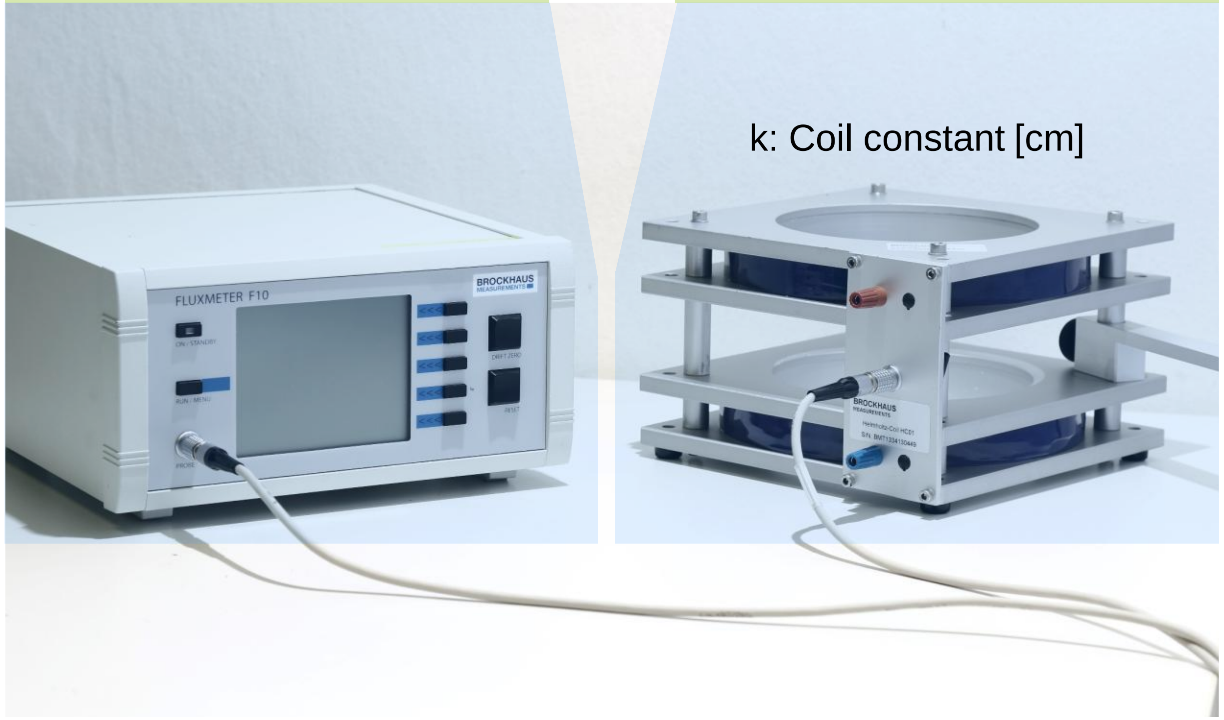
$$\int U_{ind}\, dt = -N \cdot \Phi$$

Helmholtz coil

Measuring the induced voltage

Simplified:

$$U_{ind} = -N \cdot \frac{d\Phi}{dt}$$



In a hysteresis graph apparatus, the static hysteresis curve of a permanent magnet can also be measured under the influence of temperature.

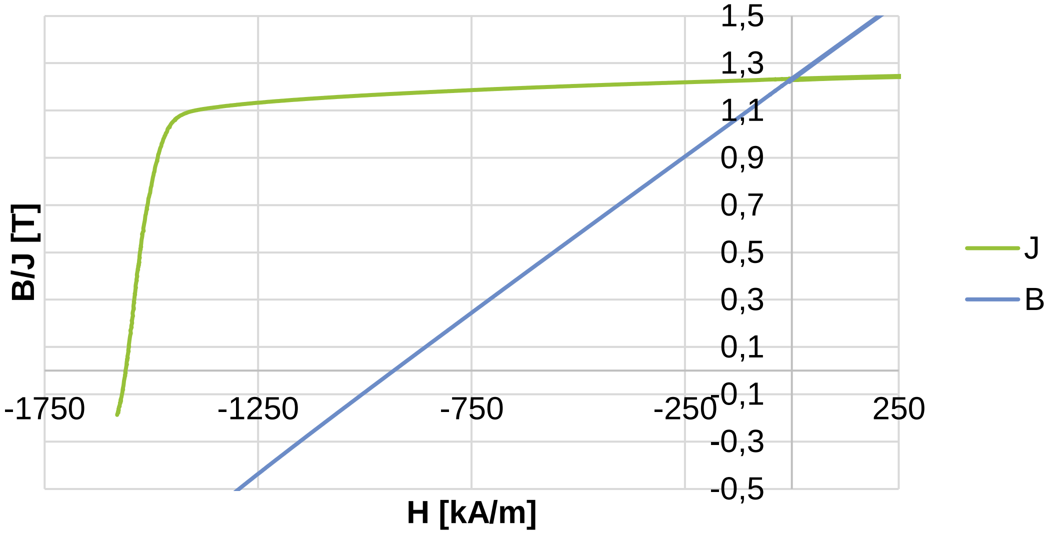
FAPS

Hysteresis graph

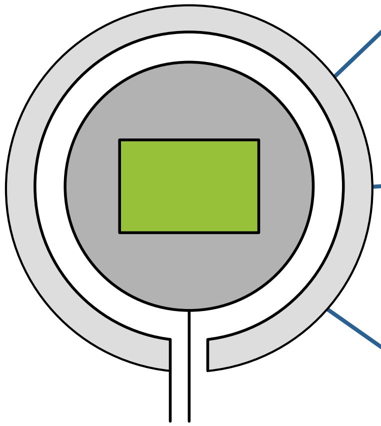
Water-cooled electromagnet

Measuring coil

Heated pole shoes



J-compensated rewinding



Magnet

Coil 1

Coil 2

- Typical demagnetization curve for an NdFeB magnet at room temperature
- Magnet is magnetically stable up to the knee point
- Variation of the characteristic values depending on the delivery quality due to
  - Position in the sintering block
  - Grain size distribution
  - Thermal history

Sensor arrays and magneto-optical sensors are suitable, to detect the field distribution on the magnetic surface.

Determination of the field distribution

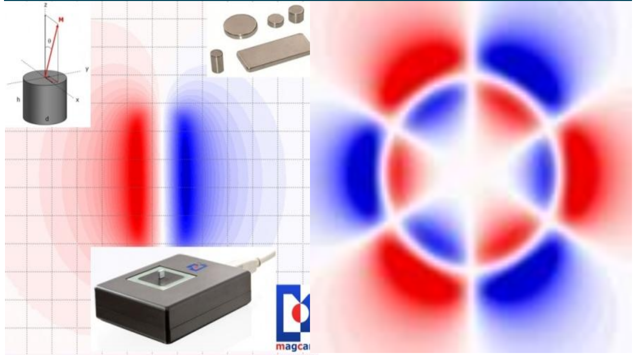
Measuring system: Magnetic field camera

Distribution of the magnetic flux density around the magnet

Measuring equipment:

- Hall sensor array
- Magneto-optical sensor
- AMR sensor array

Field distribution with Hall effect based magnetic field camera



Graphics: MagCam

Kerr effect:

- Rotation of the plane of polarization of linearly polarized light when reflected from a magnetized surface
- Angle of rotation is prop. to magnetization
- Conversion into brightness values by polarizer
- Use as magneto-optical sensor for visualization of mag. stray fields

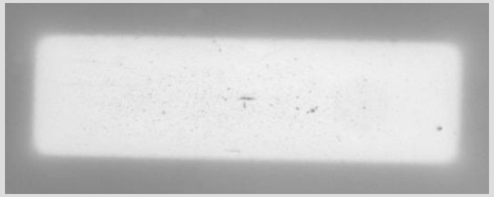
Faraday effect:

- Rotation of the plane of polarization of linearly polarized light during **transmission through a transparent magnetized volume**
- Angle of rotation is prop. to magnetization
- Dielectrics as a coating on lenses are subject to the effect
- Use as magneto-optical sensor for the visualization of mag. stray fields

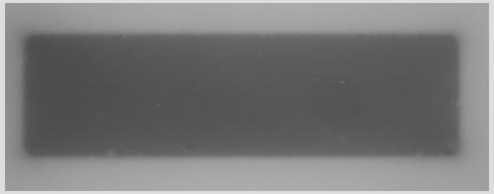
Graphics: Matesy GmbH

Magneto-optical imaging the field distribution

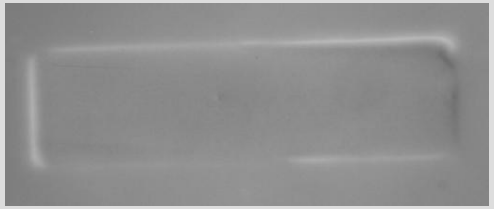
Field distribution (magn. south)



Field distribution (magn. north)



Field distribution (demagn. magnet)



In order to make the right sensor selection, the degree of automation, measuring range and requirement criteria must be checked.

Measuring equipment / criteria	Hall sensor	Fluxmeter	Hall magnetic field camera	MO magnetic field camera
Measuring principle	Selective	Selective	Surface	Surface
Measuring range	0.1 mT - 5 T	0.1 mT - 5 T	0.1 mT - 5 T	0.01 mT - 0.5 T
Resolution	High	Medium	Medium	Very high
Measuring speed (incl. data transmission)	Very high	Very high	Slow	High
Measuring capability for complex geometry	Very good	Good	Bad	Bad
Price	Low	Medium	High	High
Resilliance to interference	Good	Very good	Good	Medium

FAPS

The contents of the lecture unit can be further deepened with the following specialized literature:

Books

- Küchler, A. (2017): Hochspannungstechnik. Grundlagen - Technologie - Anwendungen. 4. Auflage. Berlin, Heidelberg: Springer Berlin Heidelberg (VDI-Buch).
- Schon, K. (2010): Stoßspannungs- und Stoßstrommesstechnik. Berlin, Heidelberg: Springer Berlin Heidelberg.
- Schon, K. (2016): Hochspannungsmesstechnik. Grundlagen – Messgeräte – Messverfahren. Wiesbaden: Springer Fachmedien Wiesbaden.
- E. Steingroever and G. Ross, *Messverfahren der Magnettechnik*. Köln: MAGNET-PHYSIK Dr. Steingroever GmbH, 2009.

Essay

Vogelsang, R; Haslimeier, R.; Fröhlich, K.; Fruth, B. (2004): Teilentladungskennwerte der elektrischen Alterung in Isolationen von Hochspannungsgeneratoren und -motoren und deren Berücksichtigung in einem TE-Monitoringsystem. In: Kindersberger J. (Hg.): Diagnostik elektrischer Betriebsmittel. Vorträge der ETG-Fachtagung am 09. und 10. März [2004] in Köln.

Internet sources

- https://www.youtube.com/@PartialDischargesGuru
- https://brockhaus.com/measurements/academy/
- https://www.lakeshore.com/products/categories/magnetic-products/gaussmeters-teslameters
- https://www.magcam.com/page/magnetic_field_measurement_systems

Prof. Dr.-Ing. Jörg Franke
Prof. Dr. rer. nat. Jens Fürst
Prof. Dr.-Ing. Florian Risch

Institute for Factory Automation and Production Systems

Friedrich-Alexander-Universität
Technische Fakultät

THANK YOU